PROJECT DESIGN PHASE

Proposed Solution Template

Date	02 October 2026	**Team ID**	SWTID-2026-9980
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value		

Proposed Solution

The project team shall provide the following information about the proposed solution.

S.No.	Parameter	Description
1	**Problem Statement (Problem to be solved)**	In systems that store sensitive records, access may need to depend on the current value of a specific field. Static role-based permissions alone may not enforce these changing conditions consistently, which can result in users viewing or modifying records they should not access.
2	**Idea / Solution Description**	Develop a script-controlled Access Control List (ACL) that evaluates a record's field value when an access request is made. The script checks the configured condition and the user's permission, then allows or denies the requested operation (such as read or write) accordingly. The rule is designed to apply consistently whenever the relevant field value changes.
3	**Novelty / Uniqueness**	The solution combines configurable field-based conditions with script-driven authorization. Instead of relying only on fixed roles, access decisions can respond to record data, supporting more granular control and easier adaptation when business rules change.
4	**Social Impact / Customer Satisfaction**	Helps protect confidential information, reduces unnecessary exposure of records, and supports more reliable handling of sensitive data. Clear, condition-based access decisions can also improve user trust and help administrators manage permissions more consistently.
5	**Business Model (Revenue Model)**	The solution can be offered as a reusable configuration or implementation service for organizations using compatible record-management platforms. Potential value may come from setup, customization, integration, maintenance, and support. Any commercial model would depend on the platform and deployment requirements.
6	**Scalability of the Solution**	The ACL logic can be extended to additional tables, fields, record types, and access operations by configuring reusable conditions. For larger deployments, rules can be documented, tested, version-controlled, and monitored to maintain performance and consistent enforcement.